

# I3C Test Report

# List of Contents

|           |                                                  |           |
|-----------|--------------------------------------------------|-----------|
| <b>1.</b> | <b>HISTORY .....</b>                             | <b>3</b>  |
| <b>2.</b> | <b>ENVIRONMENT SETUP .....</b>                   | <b>4</b>  |
|           | 2.1 ENVIRONMENT SETUP .....                      | 4         |
| <b>3.</b> | <b>I3C COMMAND EXAMPLE .....</b>                 | <b>5</b>  |
|           | 3.1 CONFIGURATION COMMAND LIST.....              | 5         |
|           | 3.1.1 Protocol Version .....                     | 5         |
|           | 3.1.2 Firmware Version .....                     | 6         |
|           | 3.1.3 Reset Notification .....                   | 7         |
|           | 3.1.4 Device Descriptor .....                    | 8         |
|           | 3.1.5 GPIO Enumeration .....                     | 10        |
|           | 3.1.6 I2C Enumeration .....                      | 11        |
|           | 3.1.7 SPI Enumeration .....                      | 12        |
|           | 3.2 GPIO COMMANDS LIST.....                      | 13        |
|           | 3.2.1 GPIO Deinit .....                          | 14        |
|           | 3.2.2 GPIO Init.....                             | 14        |
|           | 3.2.3 GPIO READ .....                            | 15        |
|           | 3.2.4 GPIO Write .....                           | 16        |
|           | 3.3 I2C COMMANDS LIST .....                      | 17        |
|           | 3.3.1 I2C Deinit .....                           | 17        |
|           | 3.3.2 I2C Init .....                             | 18        |
|           | 3.3.3 I2C Read .....                             | 19        |
|           | 3.3.4 I2C Write .....                            | 21        |
|           | 3.3.5 I2C Write Read .....                       | 22        |
|           | 3.4 SPI COMMANDS LIST .....                      | 23        |
|           | 3.4.1 SPI Deinit .....                           | 23        |
|           | 3.4.2 SPI Init .....                             | 24        |
|           | 3.4.3 SPI Read .....                             | 25        |
|           | 3.4.4 SPI Write .....                            | 26        |
|           | 3.4.5 SPI Write Read .....                       | 27        |
| <b>4.</b> | <b>I3C EXERCISER AND PROTOCOL ANALYZER .....</b> | <b>29</b> |
|           | 4.1 COMMAND(I2C READ) WAVEFORM GRAPH.....        | 29        |

# 1. History

| Version | Date       | Description                 |
|---------|------------|-----------------------------|
| V1.0    | 2023-12-14 | First version               |
| V1.1    | 2023-12-20 | Update I3C command example. |
|         |            |                             |

## 2. Environment Setup

### 2.1 Environment Setup

- Master: PGY-I3C-EX-PD I3C Exerciser and Protocol Analyzer
- Slave: EVB Board HX6538 EVB WLCSP65 and USB Debug Board

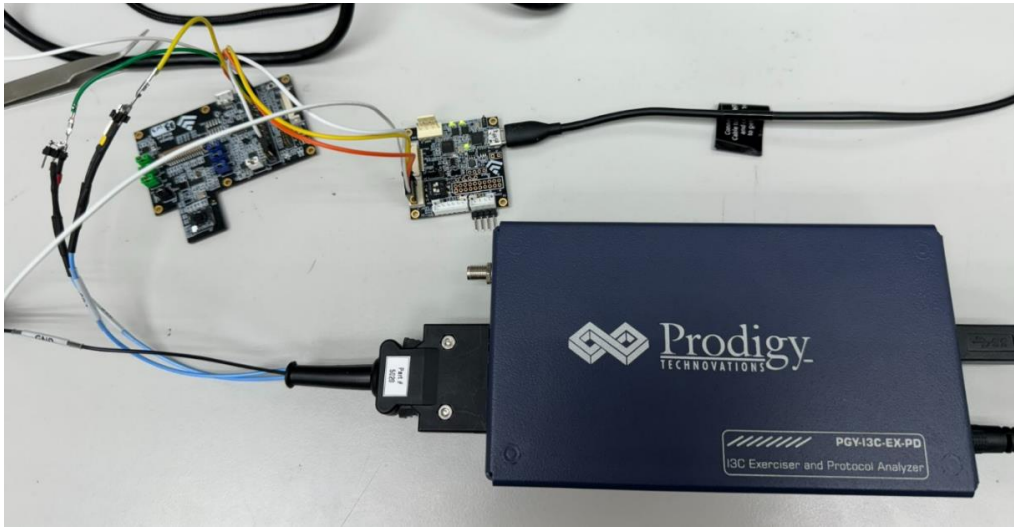


Figure 1. I3C Exerciser and Protocol Analyzer Diagram

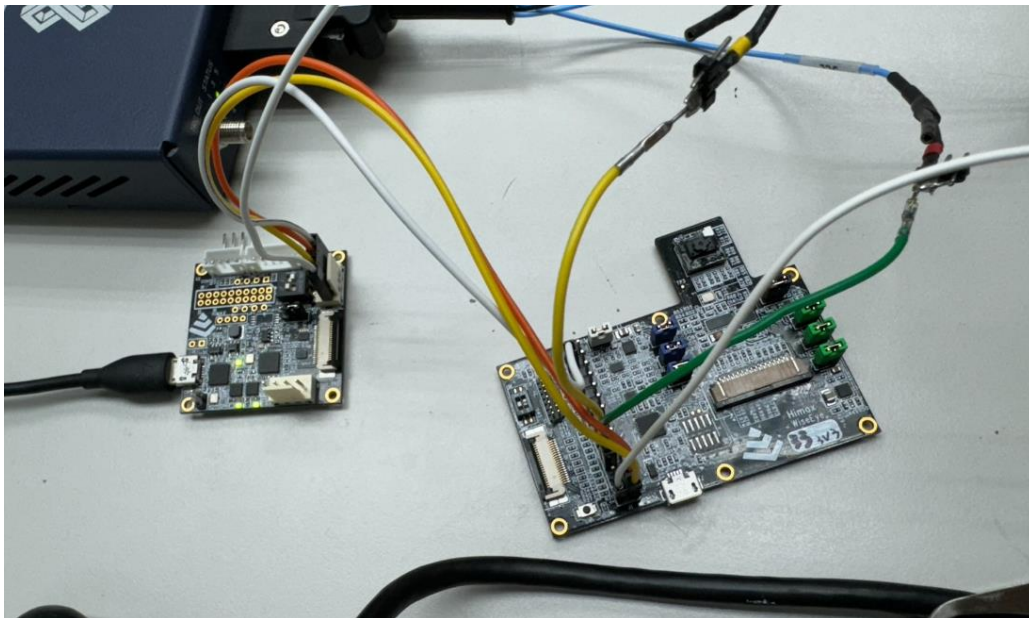


Figure 2. EVB Board and Debug Board Diagram

## 3. I3C Command Example

The host-device communication follows the protocol defined in below format.

| Type   | Command | Flags  | Payload Length | Payload              |
|--------|---------|--------|----------------|----------------------|
| 1 Byte | 1 Byte  | 1 Byte | 1 or 2 Byte(s) | (MAX – HEADER) Bytes |

Figure 3. Data Packet Format

### 3.1 Configuration Command List

| Control command    | Value       | Description                                                                                                          |
|--------------------|-------------|----------------------------------------------------------------------------------------------------------------------|
| Protocol Version   | 0x00        | Get supported Protocol Version (BCD)                                                                                 |
| Firmware Version   | 0x01        | Get Runtime Firmware Version                                                                                         |
| Reset Notification | 0x02        | Reset notification event. Query/Notify peer of the reset state                                                       |
| Reset              | 0x03        | Host directs Firmware to reset device                                                                                |
| Power State Change | 0x04        | Host directs Firmware to change power state: D0, D3, etc.                                                            |
| FwUpdate mode      | 0x05        | Host directs Firmware to reset device into FwUpdate mode.<br>This is for Discrete i3c chip option (Reserved for now) |
| Get Attributes     | 0x06        | Number of Buffers, Buffer Size for Synchronous communication, etc.                                                   |
| Reserved           | 0x07 – 0x0F | Reserved values                                                                                                      |
| GPIO Enum          | 0x10        | GPIO controller(s) enumeration                                                                                       |
| I2C Enum           | 0x11        | I2C controller(s) enumeration                                                                                        |
| SPI Enum           | 0x12        | SPI controller(s) enumeration                                                                                        |
| Reserved           | 0x13 – 0xFF | Reserved values                                                                                                      |

Figure 4. Configuration Command List

Some of control command(Reset, Power State Chang, FwUpdate mode, Get Attributes) are not define in specification.

#### 3.1.1 Protocol Version

##### Payload Format:

Bridge will request the device the supported Protocol version.

| Type | Command | Flags | Payload Length | Payload |
|------|---------|-------|----------------|---------|
| 0x01 | 0x00    | 0x05  | 0              | NA      |

The composite device will reply acknowledging the request and returning the BCD Protocol Version.

| Type | Command | Flags | Payload Length | Payload                     |
|------|---------|-------|----------------|-----------------------------|
| 0x01 | 0x00    | 0x06  | 1              | 0x20 (2.0 Protocol Version) |

### **Master(PGY-I3C-EX-PD) Request:**

- 0x01-0x00-0x05-0x00

### **Slave(WE2) Response:**

- 0x01-0x00-0x06-0x01-0x20

### **Slave(WE2) Log:**

#### **Slave Receive**

```
[I3CS_0_SDR_Rx_Handler] size:4  
i3c_slv_sdr_rx_buf[0:0x01] [1:0x00] [2:0x05] [3:0x00]  
[MSG] Type:0x1 Command:0x0 Flag:0x5 Payload_Len:0x0  
[PACKET_TYPE_CONTROL]  
[Need response]  
[PROTOCOL_VERSION]  
[dump i3c_slv_sdr_tx_buf]  
01 00 06 01 20 | ....  
[I3CS_0_SDR_Tx_Handler] size:5  
Reset TX FIFO
```

#### **Slave Transmit**

## **3.1.2 Firmware Version**

### **Payload Format:**

Bridge will request the composite device the Firmware version.

| Type | Command | Flags | Payload Length | Payload |
|------|---------|-------|----------------|---------|
| 0x01 | 0x01    | 0x05  | 0              | NA      |

The composite device will reply acknowledging the request and returning the Firmware Version in the format:

### 2.3.3. FIRMWARE VERSION

| Field | Size   | Value  | Description            |
|-------|--------|--------|------------------------|
| Major | 1 Byte | Number | Firmware major version |
| Minor | 1 Byte | Number | Firmware minor version |
| Patch | 2 Byte | Number | Firmware patch version |
| Build | 2 Byte | Number | Firmware build version |

| Type | Command | Flags | Payload Length     | Payload              |
|------|---------|-------|--------------------|----------------------|
| 0x01 | 0x01    | 0x06  | sizeof(FW_VERSION) | FW_VERSION structure |

#### Master(PGY-I3C-EX-PD) Request:

- 0x01-0x01-0x05-0x00

#### Slave(WE2) Response:

- 0x01-0x01-0x06-0x06-0x02-0x01-0x44-0x00-0x55-0x00

#### Slave(WE2) Log:

##### Slave Receive

```
[I3CS_0_SDR_Rx_Handler] size:4
i3c_slv_sdr_rx_buf[0:0x01] [1:0x01] [2:0x05] [3:0x00]
[MSG] Type:0x1 Command:0x1 Flag:0x5 Payload_Len:0x0
[PACKET_TYPE_CONTROL]
[Need response]
[FIRMWARE_VERSION]
[dump i3c_slv_sdr_tx_buf]
01 01 06 06 02 01 44 00 55 00 | .....D.U.
[I3CS_0_SDR_Tx_Handler] size:10
Reset TX FIFO
```

##### Slave Transmit

## 3.1.3 Reset Notification

#### Payload Format:

Composite Bridge requests composite device to notify once it is in reset state and ready to receive commands.

| Type | Command | Flags | Payload Length | Payload |
|------|---------|-------|----------------|---------|
| 0x01 | 0x02    | 0x05  | 0              | NA      |

The composite device will reply acknowledging the request to establish the handshake.

| Type | Command | Flags | Payload Length | Payload |
|------|---------|-------|----------------|---------|
| 0x01 | 0x02    | 0x06  | 0              | NA      |

#### **Master(PGY-I3C-EX-PD) Request:**

- 0x01-0x02-0x05-0x00

#### **Slave(WE2) Response:**

- 0x01-0x02-0x06-0x00

#### **Slave(WE2) Log:**

##### **Slave Receive**

```
[I3CS_0_SDR_Rx_Handler] size:4
i3c_slv_sdr_rx_buf[0:0x01] [1:0x02] [2:0x05] [3:0x00]
[MSG] Type:0x1 Command:0x2 Flag:0x5 Payload_Len:0x0
[PACKET_TYPE_CONTROL]
[Need response]
[RESET_NOTIFICATION]
[Handshake] Reset state and ready to receive commands
[dump i3c_slv_sdr_tx_buf]
01 02 06 00 | ....
[I3CS_0_SDR_Tx_Handler] size:4
Reset TX FIFO
```

##### **Slave Transmit**

### **3.1.4 Device Descriptor**

#### **Payload Format:**



Bridge will request the device the Device Descriptor.

| Type | Command | Flags | Payload Length | Payload |
|------|---------|-------|----------------|---------|
| 0x01 | 0x07    | 0x05  | 0              | NA      |

The composite device will reply acknowledging the request and returning the Device Descriptor.

#### Device Descriptor Structure

| Field             | Size    | Value  | Description                                    |
|-------------------|---------|--------|------------------------------------------------|
| Vendor ID         | 2 Byte  | Number | Vendor ID                                      |
| Product ID        | 2 Byte  | Number | Product ID                                     |
| Max Packet Size 0 | 1 Byte  | Number | Maximum Packet size supported for Synchronous  |
| Max Packet Size 1 | 2 Bytes | Number | Maximum Packet size supported for Asynchronous |
| Reserved          | 9 Bytes |        | Reserved for future attributes                 |
|                   |         |        |                                                |

| Type | Command | Flags | Payload Length      | Payload                     |
|------|---------|-------|---------------------|-----------------------------|
| 0x01 | 0x07    | 0x06  | sizeof(Device_Desc) | Device Descriptor structure |

#### Master(PGY-I3C-EX-PD) Request:

- 0x01-0x07-0x05-0x00

#### Slave(WE2) Response:

- 0x01-0x07-0x06-0x10-0x66-0x55-0x22-0x11-0x44-0x33-0x55-0x00-0x00-0x00-0x00-0x00-0x00-0x00-0x00

#### Slave(WE2) Log:

##### Slave Receive

```
[I3CS_0_SDR_Rx_Handler] size:4
i3c_slv_sdr_rx_buf[0:0x01] [1:0x07] [2:0x05] [3:0x00]
[MSG] Type:0x1 Command:0x7 Flag:0x5 Payload_Len:0x0
[PACKET_TYPE_CONTROL]
[Need response]
[DEVICE_DESCRIPTOR]
[dump i3c_slv_sdr_tx_buf]
01 07 06 10 66 55 22 11 44 33 55 00 00 00 00 00 | ....fU".D3U.....
00 00 00 00 | ....
[I3CS_0_SDR_Tx_Handler] size:20
Reset TX FIFO
```

##### Slave Transmit

## 3.1.5 GPIO Enumeration

### Payload Format:

| Type | Command | Flags | Payload Length | Payload |
|------|---------|-------|----------------|---------|
| 0x01 | 0x10    | 0x05  | 0              | NA      |

The composite device must report the detailed information by responding to the enumeration command with an array of N remote GPIO controllers' description in the format:

#### **GPIO BANK DESCRIPTOR**

| Field    | Size    | Value  | Description                                                                                                           |
|----------|---------|--------|-----------------------------------------------------------------------------------------------------------------------|
| BankId   | 1 Byte  | Number | GPIO(s) bank number                                                                                                   |
| PinCount | 1 Byte  | Number | Number of GPIO(s) in Pins                                                                                             |
| PinMap   | 4 Bytes | Bitmap | GPIO Pin(s) mapped:<br>Dx: Pin Value<br>0: Disabled<br>1: Enabled<br><br>Where x is the Pin number as index requested |

| Type | Command | Flags | Payload Length                 | Payload                          |
|------|---------|-------|--------------------------------|----------------------------------|
| 0x01 | 0x10    | 0x06  | N*sizeof(GPIO_BANK_DESCRIPTOR) | N*GPIO_BANK_DESCRIPTOR structure |

Where N is the number of GPIO controllers the composite device has available for the Host.

### **Master(PGY-I3C-EX-PD) Request:**

➤ 0x01-0x10-0x05-0x00

### **Slave(WE2) Response:**

➤ 0x01-0x10-0x06-0x06-0x01-0x04-0x21-0x43-0x65-0x87

### **Slave(WE2) Log:**

## Slave Receive

```
[I3CS_0_SDR_Rx_Handler] size:4
i3c_slv_sdr_rx_buf[0:0x01] [1:0x10] [2:0x05] [3:0x00]
[MSG] Type:0x1 Command:0x10 Flag:0x5 Payload_Len:0x0
[PACKET_TYPE_CONTROL]
[Need response]
[GPIO_CONTROLLER_ENUM]
[dump i3c_slv_sdr_tx_buf]
01 10 06 06 01 04 21 43 65 87 | .....!Ce.
[I3CS_0_SDR_Tx_Handler] size:10
Reset TX FIFO
```

## Slave Transmit

### 3.1.6 I2C Enumeration

#### Payload Format:

| Type | Command | Flags | Payload Length | Payload |
|------|---------|-------|----------------|---------|
| 0x01 | 0x11    | 0x05  | 0              | NA      |

#### I2C DESCRIPTOR

| Field        | Size   | Value  | Description                                                                                                                                                                                                                      |
|--------------|--------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BusId        | 1 Byte | Number | I2C bus number                                                                                                                                                                                                                   |
| Capabilities | 1 Byte | Bitmap | Controller supported capabilities:<br>D3: Address mode<br>0: 7bit address only<br>1: 10bit address<br>D1..0: Mode<br>0: Standard Mode<br>1: Fast Mode<br>2: Fast Mode+<br>3: High-speed Mode<br><br>Not listed bits are reserved |

| Type | Command | Flags | Payload Length           | Payload                    |
|------|---------|-------|--------------------------|----------------------------|
| 0x01 | 0x11    | 0x06  | N*sizeof(I2C_DESCRIPTOR) | N*I2C_DESCRIPTOR structure |

#### Master(PGY-I3C-EX-PD) Request:

➤ 0x01-0x11-0x05-0x00

#### Slave(WE2) Response:

➤ 0x01-0x11-0x06-0x04-0x00-0x01-0x01-0x01

#### Slave(WE2) Log:

#### Slave Receive

```
[I3CS_0_SDR_Rx_Handler] size:4
i3c_slv_sdr_rx_buf[0:0x01] [1:0x11] [2:0x05] [3:0x00]
[MSG] Type:0x1 Command:0x11 Flag:0x5 Payload_Len:0x0
[PACKET_TYPE_CONTROL]
[Need response]
[I2C_CONTROLLER_ENUM]
[dump i3c_slv_sdr_tx_buf]
01 11 06 04 00 01 01 01 | .....
[I3CS_0_SDR_Tx_Handler] size:8
Reset TX FIFO
```

#### Slave Transmit

### 3.1.7 SPI Enumeration

#### Payload Format:

| Type | Command | Flags | Payload Length | Payload |
|------|---------|-------|----------------|---------|
| 0x01 | 0x12    | 0x05  | 0              | NA      |

The composite device must report the detailed information by responding to the enumeration command with an array of N remote SPI controllers' description in the format:

#### SPI DESCRIPTOR

| Field        | Size   | Value  | Description                                                      |
|--------------|--------|--------|------------------------------------------------------------------|
| BusId        | 1 Byte | Number | SPI bus number                                                   |
| Capabilities | 1 Byte | Bitmap | Controller supported capabilities:<br>D3: Wire mode<br>0: 4 Wire |

|  |  |  |                                                                                                                                 |
|--|--|--|---------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | 1: 3 Wire<br>D1..0: Addressable devices<br>0: CS0<br>1: CS0/1<br>2: CS0/1/2<br>3: CS0/1/2/3<br><br>Not listed bits are reserved |
|--|--|--|---------------------------------------------------------------------------------------------------------------------------------|

| Type | Command | Flags | Payload Length           | Payload                    |
|------|---------|-------|--------------------------|----------------------------|
| 0x01 | 0x12    | 0x06  | N*sizeof(SPI_DESCRIPTOR) | N*SPI_DESCRIPTOR structure |

### Master(PGY-I3C-EX-PD) Request:

➤ 0x01-0x12-0x05-0x00

### Slave(WE2) Response:

➤ 0x01-0x12-0x06-0x02-0x00-0x03

### Slave(WE2) Log:

#### Slave Receive

```
[I3CS_0_SDR_Rx_Handler] size:4
i3c_slv_sdr_rx_buf[0:0x01] [1:0x12] [2:0x05] [3:0x00]
[MSG] Type:0x1 Command:0x12 Flag:0x5 Payload_Len:0x0
[PACKET_TYPE_CONTROL]
[Need response]
[SPI CONTROLLER ENUM]
[dump i3c_slv_sdr_tx_buf]
01 12 06 02 00 03 | .....
[I3CS_0_SDR_Tx_Handler] size:6
Reset TX FIFO
```

#### Slave Transmit

## 3.2 GPIO Commands List

| Command     | Value       | Description                                                     |
|-------------|-------------|-----------------------------------------------------------------|
| GPIO_DEINIT | 0x00        | De-initialize GPIO(s)                                           |
| GPIO_INIT   | 0x01        | Initialize GPIO(s) with the configuration passed in the payload |
| GPIO_READ   | 0x02        | Read GPIO(s) value(s)                                           |
| GPIO_WRITE  | 0x03        | Write GPIO(s) value(s)                                          |
| Reserved    | 0x04 – 0xFF | Reserved values                                                 |

Figure 5. GPIO Commands List

## 3.2.1 GPIO Deinit

### Payload Format:

#### GPIO DEINIT

| Field    | Size           | Value   | Description                                                                                 |
|----------|----------------|---------|---------------------------------------------------------------------------------------------|
| BankId   | 1 Byte         | Number  | GPIO(s) bank number                                                                         |
| Flags    | 1 Byte         | Bitmap  | Flags:<br>D0: Preserve:<br>1: Preserve Configuration<br>0: Default Configuration (Optional) |
| PinCount | 1 Byte         | Number  | Number of GPIO(s) in Pins                                                                   |
| Pins     | PinCount Bytes | Numbers | GPIO Pin(s) array                                                                           |

Host could request to de-initialize 1 or more GPIO Pin(s):

| Type | Command | Flags                 | Payload Length      | Payload     |
|------|---------|-----------------------|---------------------|-------------|
| 0x03 | 0x00    | <a href="#">FLAGS</a> | sizeof(GPIO_DEINIT) | GPIO_DEINIT |

When requested, the expected response from composite device:

| Type | Command | Flags                 | Payload Length      | Payload     |
|------|---------|-----------------------|---------------------|-------------|
| 0x03 | 0x00    | <a href="#">FLAGS</a> | sizeof(GPIO_DEINIT) | GPIO_DEINIT |

### Master(PGY-I3C-EX-PD) Request:

- 0x03-0x00-0x05-0x06-0x00-0x00-0x03-0x00-0x01-0x02

### Slave(WE2) Response:

- 0x03-0x01-0x06-0x06-0x00-0x00-0x03-0x00-0x01-0x02

### Slave(WE2) Log:

#### Slave Receive

```
[I3CS_0_SDR_Rx_Handler] size:10
i3c_slv_sdr_rx_buf[0:0x03] [1:0x00] [2:0x05] [3:0x06] [4:0x00] [5:0x00] [6:0x03] [7:0x00] [8:0x01] [9:0x02]
[MSG] Type:0x03 Command:0x00 Flag:0x05 Payload_Len:0x06
[PACKET_TYPE_GPIO]
[Need response]
[GPIO_DEINIT]
Dump request payload
00 00 03 00 01 02 | .....
[dump i3c_slv_sdr_tx_buf]
03 01 06 06 00 00 03 00 01 02 | .....
[I3CS_0_SDR_Tx_Handler] size:10
Reset TX FIFO
```

#### Slave Transmit

## 3.2.2 GPIO Init

### Payload Format:

## GPIO INIT

| Field    | Size           | Value   | Description                                                                                                                                              |
|----------|----------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| BankId   | 1 Byte         | Number  | GPIO(s) bank number                                                                                                                                      |
| Config   | 1 Byte         | Bitmap  | Config:<br>D3..2: PinConfig<br>0: Default<br>1: Pullup<br>2: Pulldown<br>3: None<br>D1..0: PinMode<br>0: Invalid<br>1: Input<br>2: Output<br>3: Reserved |
| PinCount | 1 Byte         | Number  | Number of GPIO(s) in Pins                                                                                                                                |
| Pins     | PinCount Bytes | Numbers | GPIO Pin(s) array                                                                                                                                        |

Host could request to initialize 1 or more GPIO Pin(s):

| Type | Command | Flags                 | Payload Length    | Payload   |
|------|---------|-----------------------|-------------------|-----------|
| 0x03 | 0x01    | <a href="#">FLAGS</a> | sizeof(GPIO_INIT) | GPIO_INIT |

When requested, the expected response from composite device:

| Type | Command | Flags                 | Payload Length    | Payload   |
|------|---------|-----------------------|-------------------|-----------|
| 0x03 | 0x01    | <a href="#">FLAGS</a> | sizeof(GPIO_INIT) | GPIO_INIT |

### Master(PGY-I3C-EX-PD) Request:

- 0x03-0x01-0x05-0x05-0x00-0x02-0x02-0x01-0x02

### Slave(WE2) Response:

- 0x03-0x01-0x06-0x05-0x00-0x02-0x02-0x01-0x02

### Slave(WE2) Log:

#### Slave Receive

```
[I3CS_0_SDR_Rx_Handler] size:9
i3c_slv_sdr_rx_buf[0:0x03] [1:0x01] [2:0x05] [3:0x05] [4:0x00] [5:0x02] [6:0x02] [7:0x01] [8:0x02]
[MSG] Type:0x3 Command:0x1 Flag:0x5 Payload_Len:0x5
[PACKET_TYPE_GPIO]
[Need response]
[GPIO_INIT]
Dump request payload
00 02 02 01 02 |
[dump i3c_slv_sdr_tx_buf]
03 01 06 05 00 02 02 01 02 | .....
[I3CS_0_SDR_Tx_Handler] size:9
Reset TX FIFO
```

#### Slave Transmit

## 3.2.3 GPIO READ

### Payload Format:

### GPIO READ

| Field    | Size           | Value   | Description                                                                                                                                                              |
|----------|----------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BankId   | 1 Byte         | Number  | GPIO(s) bank number                                                                                                                                                      |
| PinCount | 1 Byte         | Number  | Number of GPIO(s) in Pins                                                                                                                                                |
| Pins     | PinCount Bytes | Numbers | GPIO Pin(s) array                                                                                                                                                        |
| Value    | 4 Bytes        | Bitmap  | GPIO Pin(s) value(s) mapped:<br>Dx: Pin Value<br>0: Low<br>1: High<br><br>Where x is the Pin number as index requested<br><b>*Note: Value is not sent in the request</b> |

The host could request to read 1 or more GPIO Pin(s):

| Type | Command | Flags                 | Payload Length        | Payload   |
|------|---------|-----------------------|-----------------------|-----------|
| 0x03 | 0x02    | <a href="#">FLAGS</a> | sizeof(GPIO_READ) - 4 | GPIO_READ |

When requested, the expected response from composite device:

| Type | Command | Flags                 | Payload Length    | Payload   |
|------|---------|-----------------------|-------------------|-----------|
| 0x03 | 0x02    | <a href="#">FLAGS</a> | sizeof(GPIO_READ) | GPIO_READ |

### Master(PGY-I3C-EX-PD) Request:

- 0x03-0x02-0x05-0x03-0x00-0x01-0x02

### Slave(WE2) Response:

- 0x03-0x02-0x06-0x07-0x00-0x01-0x02-0x44-0x33-0x22-0x11

### Slave(WE2) Log:

#### Slave Receive

```
[I3CS_0_SDR_Rx_Handler] size:7
i3c_slv_sdr_rx_buf[0:0x03] [1:0x02] [2:0x05] [3:0x03] [4:0x00] [5:0x01] [6:0x02]
[MSG] Type:0x3 Command:0x2 Flag:0x5 Payload_Len:0x3
[PACKET_TYPE_GPIO]
[Need response]
[GPIO_READ]
Dump request payload
00 01 02 | ...
GPIO_READ PinCount:1 Array
[0]0x2
[dump i3c_slv_sdr_tx_buf]
03 02 06 07 00 01 02 44 33 22 11 | .....D3".
[I3CS_0_SDR_Tx_Handler] size:11
Reset TX FIFO
Slave Transmit
```

## 3.2.4 GPIO Write

### Payload Format:



### Master(PGY-I3C-EX-PD) Request:

- 0x03-0x03-0x05-0x07-0x00-0x01-0x01-0x02-0x00-0x00-0x00

### Slave(WE2) Response:

- 0x03-0x03-0x06-0x07-0x00-0x01-0x01-0x02-0x00-0x00-0x00

### Slave(WE2) Log:

#### Slave Receive

```
[I3CS_0_SDR_Rx_Handler] size:11
i3c_slv_sdr_rx_buf[0:0x03] [1:0x03] [2:0x05] [3:0x07] [4:0x00] [5:0x01] [6:0x01] [7:0x02] [8:0x00] [9:0x00] [10:0x00]
[MSG] Type:0x3 Command:0x3 Flag:0x3 Payload_Len:0x7
[PACKET_TYPE_GPIO]
[Need response]
[GPIO_WRITE]
Dump request payload
00 01 01 02 00 00 00 | .....
GPIO_WRITE PinCount:1 Array
[0]0x1
write_value:0x2
[dump i3c_slv_sdr_tx_buf]
03 03 06 07 00 01 01 02 00 00 00 | .....
[I3CS_0_SDR_Tx_Handler] size:11
Reset TX FIFO
```

#### Slave Transmit

## 3.3 I2C Commands List

| Command       | Value       | Description                             |
|---------------|-------------|-----------------------------------------|
| I2C_DEINIT    | 0x00        | De-initialize target device             |
| I2C_INIT      | 0x01        | Initialize target device                |
| I2C_READ      | 0x02        | Read from a target device               |
| I2C_WRITE     | 0x03        | Write to a target device                |
| I2C_WRITEREAD | 0x04        | Write then read to/from a target device |
| Reserved      | 0x05 – 0xFF | Reserved values                         |

Figure 6. I2C Commands List

### 3.3.1 I2C Deinit

#### Payload Format:

| Field  | Size    | Value  | Description                                                                                                                                                                                                                                                       |
|--------|---------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BusId  | 1 Byte  | Number | I2C Controller number                                                                                                                                                                                                                                             |
| Config | 2 Bytes | Bitmap | Config:<br>D15: Address mode<br>0: 7-bit address mode<br>1: 10-bit address mode<br>D9..7: In 10-bit address mode, this represents the highest 3 bits of the address.<br>D6..0: Lowest 7-bit address. In 7-bit address mode, this represents the complete address. |

The host issues the request command:

| Type | Command | Flags                 | Payload Length     | Payload    |
|------|---------|-----------------------|--------------------|------------|
| 0x04 | 0x00    | <a href="#">FLAGS</a> | sizeof(I2C_DEINIT) | I2C_DEINIT |

When requested, the expected response from composite device:

| Type | Command | Flags                 | Payload Length     | Payload    |
|------|---------|-----------------------|--------------------|------------|
| 0x04 | 0x00    | <a href="#">FLAGS</a> | sizeof(I2C_DEINIT) | I2C_DEINIT |

### **Master(PGY-I3C-EX-PD) Request:**

- 0x04-0x00-0x05-0x03-0x01-0x25-0x00

### **Slave(WE2) Response:**

- 0x04-0x00-0x06-0x03-0x01-0x25-0x00

### **Slave(WE2) Log:**

#### **Slave Receive**

```
[I3CS_0_SDR_Rx_Handler] size:7
i3c_slv_sdr_rx_buf[0:0x04] [1:0x00] [2:0x05] [3:0x03] [4:0x01] [5:0x25] [6:0x00]
[MSG] Type:0x4 Command:0x0 Flag:0x5 Payload_Len:0x3
[PACKET_TYPE_I2C]
[Need response]
[I2C_DEINIT]
Dump request payload
01 25 00 | %
[dump i3c_slv_sdr_tx_buf]
04 00 06 03 01 25 00 | .....%.
[I3CS_0_SDR_Tx_Handler] size:7
Reset TX FIFO
```

#### **Slave Transmit**

## **3.3.2 I2C Init**

### **Payload Format:**

| Field  | Size    | Value  | Description                                                                                                                                                                                                                                                       |
|--------|---------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BusId  | 1 Byte  | Number | I2C Controller number                                                                                                                                                                                                                                             |
| Config | 2 Bytes | Bitmap | Config:<br>D15: Address mode<br>0: 7-bit address mode<br>1: 10-bit address mode<br>D9..7: In 10-bit address mode, this represents the highest 3 bits of the address.<br>D6..0: Lowest 7-bit address. In 7-bit address mode, this represents the complete address. |
| Speed  | 4 Bytes | Number | Bus speed in Hertz                                                                                                                                                                                                                                                |

The host issues the request command:

| Type | Command | Flags                 | Payload Length   | Payload  |
|------|---------|-----------------------|------------------|----------|
| 0x04 | 0x01    | <a href="#">FLAGS</a> | sizeof(I2C_INIT) | I2C_INIT |

When requested, the expected response from composite device:

| Type | Command | Flags                 | Payload Length   | Payload  |
|------|---------|-----------------------|------------------|----------|
| 0x04 | 0x01    | <a href="#">FLAGS</a> | sizeof(I2C_INIT) | I2C_INIT |

### **Master(PGY-I3C-EX-PD) Request:**

- 0x04-0x01-0x05-0x07-0x01-0x25-0x00-0x80-0x1A-0x06-0x00

### **Slave(WE2) Response:**

- 0x04-0x01-0x06-0x07-0x01-0x25-0x00-0x80-0x1A-0x06-0x00

### **Slave(WE2) Log:**

#### **Slave Receive**

```
[I3CS_0_SDR_Rx_Handler] size:11
i3c_slv_sdr_rx_buf[0:0x04] [1:0x01] [2:0x05] [3:0x07] [4:0x01] [5:0x25] [6:0x00] [7:0x80] [8:0x1A] [9:0x06] [10:0x00]
[MSG] Type:0x4 Command:0x1 Flag:0x5 Payload_Len:0x7
[PACKET_TYPE_I2C]
[Need response]
[I2C_INIT]
Dump request payload
01 25 00 80 1A 06 00 | %
[dump i3c_slv_sdr_tx_buf]
04 01 06 07 01 25 00 80 1A 06 00 | .....%.....
[I3CS_0_SDR_Tx_Handler] size:11
Reset TX FIFO
```

#### **Slave Transmit**

## **3.3.3 I2C Read**

### **Payload Format:**

## I2C READ

| Field  | Size       | Value   | Description                                                                                                                                                                                                                                                       |
|--------|------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BusId  | 1 Byte     | Number  | I2C controller number                                                                                                                                                                                                                                             |
| Config | 2 Bytes    | Bitmap  | Config:<br>D15: Address mode<br>0: 7-bit address mode<br>1: 10-bit address mode<br>D9..7: In 10-bit address mode, this represents the highest 3 bits of the address.<br>D6..0: Lowest 7-bit address. In 7-bit address mode, this represents the complete address. |
| Size   | 2 Bytes    | Number  | Number of bytes to read                                                                                                                                                                                                                                           |
| Data   | Size Bytes | Numbers | Array of bytes read from device.<br>Note: Just returned by the device.                                                                                                                                                                                            |

Host issues the request command:

| Type | Command | Flags                 | Payload Length          | Payload  |
|------|---------|-----------------------|-------------------------|----------|
| 0x04 | 0x02    | <a href="#">FLAGS</a> | sizeof(I2C_READ) - Size | I2C_READ |

When requested, the expected response from composite device:

| Type | Command | Flags                 | Payload Length   | Payload  |
|------|---------|-----------------------|------------------|----------|
| 0x04 | 0x02    | <a href="#">FLAGS</a> | sizeof(I2C_READ) | I2C_READ |

## Master(PGY-I3C-EX-PD) Request:

- 0x04-0x02-0x05-0x05-0x01-0x00-0x01-0x04-0x00

## Slave(WE2) Response:

- 0x04-0x02-0x06-0x09-0x01-0x00-0x01-0x04-0x00-0x0C-0x00-0x38-0x85

## Slave(WE2) Log:

### Slave Receive

```
[I3CS_0_SDR_Rx_Handler] size:9
i3c_slv_sdr_rx_buf[0:0x04] [1:0x02] [2:0x05] [3:0x05] [4:0x01] [5:0x00] [6:0x01] [7:0x04] [8:0x00]
[MSG] Type:0x4 Command:0x2 Flag:0x5 Payload_Len:0x5
[PACKET_TYPE_I2C]
[Need response]
[I2C_READ]
Dump request payload
01 00 01 04 00 | .....
BusID:1 Addr_Lowest:0 Addr_Highest:2 AddressMode:0 ReadSize:4
reg_data:0x8538000C
[dump i3c_slv_sdr_tx_buf]
04 02 06 09 01 00 01 04 00 0C 00 38 85 | .....8.
[I3CS_0_SDR_Tx_Handler] size:13
Reset TX FIFO
```

### Slave Transmit

## 3.3.4 I2C Write

### Payload Format:

| Field  | Size       | Value   | Description                                                                                                                                                                                                                                                       |
|--------|------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BusId  | 1 Byte     | Number  | I2C controller number                                                                                                                                                                                                                                             |
| Config | 2 Bytes    | Bitmap  | Config:<br>D15: Address mode<br>0: 7-bit address mode<br>1: 10-bit address mode<br>D9..7: In 10-bit address mode, this represents the highest 3 bits of the address.<br>D6..0: Lowest 7-bit address. In 7-bit address mode, this represents the complete address. |
| Size   | 2 Bytes    | Number  | Number of bytes to write/written                                                                                                                                                                                                                                  |
| Data   | Size Bytes | Numbers | Array of bytes to write.<br>Note: Just sent by the host.                                                                                                                                                                                                          |

Host issues the request command:

| Type | Command | Flags                 | Payload Length    | Payload   |
|------|---------|-----------------------|-------------------|-----------|
| 0x04 | 0x03    | <a href="#">FLAGS</a> | sizeof(I2C_WRITE) | I2C_WRITE |

When requested, the expected response from composite device:

| Type | Command | Flags                 | Payload Length           | Payload   |
|------|---------|-----------------------|--------------------------|-----------|
| 0x04 | 0x03    | <a href="#">FLAGS</a> | sizeof(I2C_WRITE) - Size | I2C_WRITE |

### Master(PGY-I3C-EX-PD) Request:

- 0x04-0x03-0x05-0x0F-0x02-0x01-0x00-0x0A-0x00-0x01-0x02-0x03-0x04-0x05-0x06-0x07-0x08-0x09-0x0A

### Slave(WE2) Response:

- 0x04-0x03-0x06-0x05-0x02-0x01-0x00-0x0A-0x00

### Slave(WE2) Log:

#### Slave Receive

```
[I3CS_0_SDR_Rx_Handler] size:19
i3c_slv_sdr_rx_buf[0:0x04] [1:0x03] [2:0x05] [3:0x0F] [4:0x02] [5:0x01] [6:0x00] [7:0x0A] [8:0x00] [9:0x01]
[10:0x02] [11:0x03] [12:0x04] [13:0x05] [14:0x06] [15:0x07] [16:0x08] [17:0x09] [18:0x0A]
[MSG] type:0x4 Command:0x3 Flag:0x5 Payload_Len:0x1
[PACKET_TYPE_I2C]
[Need response]
[I2C_WRITE]
Dump request payload
02 01 00 0A 00 01 02 03 04 05 06 07 08 09 0A | .....
BusID:2 Addr_Lowest:1 Addr_Highest:0 AddressMode:0 ReadSize:10
[dump i3c_slv_sdr_tx_buf]
04 03 06 05 02 01 00 0A 00 | .....
[I3CS_0_SDR_Tx_Handler] size:9
Reset TX FIFO
```

#### Slave Transmit

## 3.3.5 I2C Write Read

### Payload Format:

| Field     | Size            | Value   | Description                                                                                                                                                                    |
|-----------|-----------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BusId     | 1 Byte          | Number  | I2C controller number                                                                                                                                                          |
| Config    | 2 Bytes         | Bitmap  | Config:<br>D15: Address mode<br>0: 7-bit address mode<br>1: 10-bit address mode                                                                                                |
|           |                 |         | D9..7: In 10-bit address mode, this represents the highest 3 bits of the address.<br>D6..0: Lowest 7-bit address. In 7-bit address mode, this represents the complete address. |
| WriteSize | 2 Bytes         | Number  | Number of bytes to write                                                                                                                                                       |
| ReadSize  | 2 Bytes         | Number  | Number of bytes to read                                                                                                                                                        |
| Data      | WriteSize Bytes | Numbers | Array of bytes to write.                                                                                                                                                       |

Host issues the request command:

| Type | Command | Flags                 | Payload Length        | Payload       |
|------|---------|-----------------------|-----------------------|---------------|
| 0x04 | 0x04    | <a href="#">FLAGS</a> | sizeof(I2C_WRITEREAD) | I2C_WRITEREAD |

From the device the format would be similar:

| Field     | Size           | Value   | Description                                                                                                                                                                                                                                                       |
|-----------|----------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BusId     | 1 Byte         | Number  | I2C controller number                                                                                                                                                                                                                                             |
| Config    | 2 Bytes        | Bitmap  | Config:<br>D15: Address mode<br>0: 7-bit address mode<br>1: 10-bit address mode<br>D9..7: In 10-bit address mode, this represents the highest 3 bits of the address.<br>D6..0: Lowest 7-bit address. In 7-bit address mode, this represents the complete address. |
| WriteSize | 2 Bytes        | Number  | Number of bytes written                                                                                                                                                                                                                                           |
| ReadSize  | 2 Bytes        | Number  | Number of bytes read                                                                                                                                                                                                                                              |
| Data      | ReadSize Bytes | Numbers | Array of bytes read.                                                                                                                                                                                                                                              |

When requested, the expected response from composite device:

| Type | Command | Flags                 | Payload Length        | Payload       |
|------|---------|-----------------------|-----------------------|---------------|
| 0x04 | 0x04    | <a href="#">FLAGS</a> | sizeof(I2C_WRITEREAD) | I2C_WRITEREAD |

### Master(PGY-I3C-EX-PD) Request:

- 0x04-0x04-0x05-0x09-0x00-0x25-0x00-0x02-0x00-0x04-0x00-0x22-0x66

### Slave(WE2) Response:

- 0x04-0x04-0x06-0x0B-0x00-0x25-0x00-0x02-0x00-0x04-0x00-0x0C-0x00-0x38-0x85

## Slave(WE2) Log:

### Slave Receive

```
[I3CS_0_SDR_Rx_Handler] size:13
i3c_slv_sdr_rx_buf[0:0x04] [1:0x04] [2:0x05] [3:0x09] [4:0x00] [5:0x25] [6:0x00] [7:0x02] [8:0x00] [9:0x04] [10:0x00] [11:0x22] [12:0x66]
[MSG] Type:0x4 Command:0x4 Flag:0x5 Payload_Len:0x3
[PACKET_TYPE_I2C]
[Need response]
[I2C_WRITEREAD]
Dump request payload
BusID:0 Addr_Lowest:37 Addr_Highest:0 AddressMode:0 WriteSize:2 ReadSize:4
00 25 00 02 00 04 00 22 66 | %.5s" f
reg_data:0x0530000C
[dump i3c_slv_sdr_tx_buf]
04 04 06 00 00 25 00 02 00 04 00 0C 00 38 85 | .....8.
[I3CS_0_SDR_Tx_Handler] size:15
Reset TX FIFO
```

### Slave Transmit

## 3.4 SPI Commands List

| Command    | Value | Description                 |
|------------|-------|-----------------------------|
| SPI_DEINIT | 0x00  | De-initialize target device |
| SPI_INIT   | 0x01  | Initialize target device    |
| SPI_READ   | 0x02  | Read from a target device   |

|               |             |                                         |
|---------------|-------------|-----------------------------------------|
| SPI_WRITE     | 0x03        | Write to a target device                |
| SPI_WRITEREAD | 0x04        | Write then read to/from a target device |
| Reserved      | 0x05 – 0xFF | Reserved values                         |

Figure 7. SPI Commands List

### 3.4.1 SPI Deinit

#### Payload Format:

### SPI DEINIT

| Field  | Size    | Value  | Description                                                                                     |
|--------|---------|--------|-------------------------------------------------------------------------------------------------|
| BusId  | 1 Byte  | Number | SPI Controller number                                                                           |
| Config | 1 Bytes | Bitmap | Config:<br>D7: Wire mode<br>0: 4 Wire mode<br>1: 3 Wire mode<br>D1..0: Chip Select 0, 1, 2 or 3 |

The host issues the request command:

| Type | Command | Flags                 | Payload Length     | Payload    |
|------|---------|-----------------------|--------------------|------------|
| 0x05 | 0x00    | <a href="#">FLAGS</a> | sizeof(SPI_DEINIT) | SPI_DEINIT |

When requested, the expected response from composite device:

| Type | Command | Flags                 | Payload Length     | Payload    |
|------|---------|-----------------------|--------------------|------------|
| 0x05 | 0x00    | <a href="#">FLAGS</a> | sizeof(SPI_DEINIT) | SPI_DEINIT |

### Master(PGY-I3C-EX-PD) Request:

- 0x05-0x00-0x05-0x02-0x00-0x00

### Slave(WE2) Response:

- 0x05-0x00-0x06-0x02-0x00-0x00

### Slave(WE2) Log:

#### Slave Receive

```
[I3CS_0_SDR_Rx_Handler] size:6
i3c_slv_sdr_rx_buf[0:0x05] [1:0x00] [2:0x05] [3:0x02] [4:0x00] [5:0x00]
[MSG] type:0x5 Command:0x00 Flag:0x5 Payload_Len:0x2
[PACKET_TYPE_SPI]
[Need response]
[SPI_DEINIT]
Dump request payload
00 00 | ..
[dump i3c_slv_sdr_tx_buf]
05 00 06 02 00 00 | .....
[I3CS_0_SDR_Tx_Handler] size:6
Reset TX FIFO
```

#### Slave Transmit

## 3.4.2 SPI Init

### Payload Format:



## SPI INIT

| Field  | Size    | Value  | Description                                                                                     |
|--------|---------|--------|-------------------------------------------------------------------------------------------------|
| BusId  | 1 Byte  | Number | SPI Controller number                                                                           |
| Config | 1 Bytes | Bitmap | Config:<br>D7: Wire mode<br>0: 4 Wire mode<br>1: 3 Wire mode<br>D1..0: Chip Select 0, 1, 2 or 3 |
| Speed  | 4 Bytes | Number | Bus speed in Hertz                                                                              |

The host issues the request command:

| Type | Command | Flags                 | Payload Length   | Payload  |
|------|---------|-----------------------|------------------|----------|
| 0x05 | 0x01    | <a href="#">FLAGS</a> | sizeof(SPI_INIT) | SPI_INIT |

When requested, the expected response from composite device:

| Type | Command | Flags                 | Payload Length   | Payload  |
|------|---------|-----------------------|------------------|----------|
| 0x05 | 0x01    | <a href="#">FLAGS</a> | sizeof(SPI_INIT) | SPI_INIT |

### Master(PGY-I3C-EX-PD) Request:

- 0x05-0x01-0x05-0x06-0x00-0x00-0x80-0x96-0x98-0x00

### Slave(WE2) Response:

- 0x05-0x01-0x06-0x06-0x00-0x00-0x80-0x96-0x98-0x00

### Slave(WE2) Log:

#### Slave Receive

```
[I3CS_0_SDR_Rx_Handler] size:10
i3c_slv_sdr_rx_buf[0:0x05] [1:0x01] [2:0x05] [3:0x06] [4:0x00] [5:0x00] [6:0x80] [7:0x96] [8:0x98] [9:0x00]
[MSG] Type:0x5 Command:0x1 Flag:0x5 Payload_Len:0x6
[PACKET_TYPE_SPI]
[Need response]
[SPI_INIT]
Dump request payload
00 00 80 96 98 00 | .....
[dump i3c_slv_sdr_tx_buf]
05 01 06 06 00 00 80 96 98 00 | .....
[I3CS_0_SDR_Tx_Handler] size:10
Reset TX FIFO
```

#### Slave Transmit

## 3.4.3 SPI Read

### Payload Format:

### SPI READ

| Field  | Size       | Value   | Description                                                                                     |
|--------|------------|---------|-------------------------------------------------------------------------------------------------|
| BusId  | 1 Byte     | Number  | SPI controller number                                                                           |
| Config | 1 Bytes    | Bitmap  | Config:<br>D7: Wire mode<br>0: 4 Wire mode<br>1: 3 Wire mode<br>D1..0: Chip Select 0, 1, 2 or 3 |
| Size   | 2 Bytes    | Number  | Number of bytes to read                                                                         |
| Data   | Size Bytes | Numbers | Array of bytes read from device.<br>Note: Just returned by the device.                          |

Host issues the request command:

| Type | Command | Flags                 | Payload Length          | Payload  |
|------|---------|-----------------------|-------------------------|----------|
| 0x05 | 0x02    | <a href="#">FLAGS</a> | sizeof(SPI_READ) - Size | SPI_READ |

When requested, the expected response from composite device:

| Type | Command | Flags                 | Payload Length   | Payload  |
|------|---------|-----------------------|------------------|----------|
| 0x05 | 0x02    | <a href="#">FLAGS</a> | sizeof(SPI_READ) | SPI_READ |

### Master(PGY-I3C-EX-PD) Request:

- 0x05-0x02-0x05-0x04-0x00-0x00-0x04-0x00

### Slave(WE2) Response:

- 0x05-0x02-0x06-0x08-0x00-0x00-0x04-0x00-0x11-0x22-0x33-0x44

### Slave(WE2) Log:

#### Slave Receive

```
[I3CS_0_SDR_Rx_Handler] size:8
i3c_slv_sdr_rx_buf[0:0x05] [1:0x02] [2:0x05] [3:0x04] [4:0x00] [5:0x00] [6:0x04] [7:0x00]
[MSG] Type:0x5 Command:0x2 Flag:0x5 Payload_Len:0x4
[PACKET_TYPE_SPI]
[Need response]
[SPI_READ]
Dump request payload
00 00 04 00 | .....
BusID:0 CS_Select:0 WireMode:0 ReadSize:4
[dump i3c_slv_sdr_tx_buf]
05 02 06 08 00 00 04 00 11 22 33 44 | ..... "3D
[I3CS_0_SDR_Tx_Handler] size:12
Set TX FIFO
```

#### Slave Transmit

## 3.4.4 SPI Write

### Payload Format:

## SPI WRITE

| Field  | Size       | Value   | Description                                                                                     |
|--------|------------|---------|-------------------------------------------------------------------------------------------------|
| BusId  | 1 Byte     | Number  | SPI controller number                                                                           |
| Config | 1 Bytes    | Bitmap  | Config:<br>D7: Wire mode<br>0: 4 Wire mode<br>1: 3 Wire mode<br>D1..0: Chip Select 0, 1, 2 or 3 |
| Size   | 2 Bytes    | Number  | Number of bytes to write/written                                                                |
| Data   | Size Bytes | Numbers | Array of bytes to write.<br>Note: Just sent by the host.                                        |

Host issues the request command:

| Type | Command | Flags                 | Payload Length    | Payload   |
|------|---------|-----------------------|-------------------|-----------|
| 0x05 | 0x03    | <a href="#">FLAGS</a> | sizeof(SPI_WRITE) | SPI_WRITE |

When requested, the expected response from composite device:

| Type | Command | Flags                 | Payload Length           | Payload   |
|------|---------|-----------------------|--------------------------|-----------|
| 0x05 | 0x03    | <a href="#">FLAGS</a> | sizeof(SPI_WRITE) - Size | SPI_WRITE |

### Master(PGY-I3C-EX-PD) Request:

- 0x05-0x03-0x05-0x08-0x00-0x00-0x04-0x00-0x44-0x33-0x22-0x11

### Slave(WE2) Response:

- 0x05-0x03-0x06-0x04-0x00-0x00-0x04-0x00

### Slave(WE2) Log:

#### Slave Receive

```
[I3CS_0_SDR_Rx_Handler] size:12
i3c_slv_sdr_rx_buf[0:0x05] [1:0x03] [2:0x05] [3:0x08] [4:0x00] [5:0x00] [6:0x04] [7:0x00] [8:0x44] [9:0x33] [10:0x22] [11:0x11]
[MSG] Type:0x5 Command:0x3 Flag:0x5 Payload_Len:0x8
[PACKET_TYPE_SPI]
[Need response]
[SPI_WRITE]
Dump request payload
00 00 04 00 44 33 22 11 | ....D3".
BusID:0 CS_Select:0 WireMode:0 WriteSize:4
[dump i3c_slv_sdr_tx_buf]
05 03 06 04 00 00 04 00 | .....
[I3CS_0_SDR_Tx_Handler] size:8
Reset TX FIFO
```

#### Slave Transmit

## 3.4.5 SPI Write Read

### Payload Format:

## SPI WRITEREAD

| Field     | Size            | Value   | Description                                                                                     |
|-----------|-----------------|---------|-------------------------------------------------------------------------------------------------|
| BusId     | 1 Byte          | Number  | SPI controller number                                                                           |
| Config    | 1 Bytes         | Bitmap  | Config:<br>D7: Wire mode<br>0: 4 Wire mode<br>1: 3 Wire mode<br>D1..0: Chip Select 0, 1, 2 or 3 |
| WriteSize | 2 Bytes         | Number  | Number of bytes to write                                                                        |
| ReadSize  | 2 Bytes         | Number  | Number of bytes to read                                                                         |
| Data      | WriteSize Bytes | Numbers | Array of bytes to write.                                                                        |

Host issues the request command:

| Type | Command | Flags                 | Payload Length        | Payload       |
|------|---------|-----------------------|-----------------------|---------------|
| 0x05 | 0x04    | <a href="#">FLAGS</a> | sizeof(SPI_WRITEREAD) | SPI_WRITEREAD |

When requested, the expected response from composite device:

| Type | Command | Flags                 | Payload Length        | Payload       |
|------|---------|-----------------------|-----------------------|---------------|
| 0x05 | 0x04    | <a href="#">FLAGS</a> | sizeof(SPI_WRITEREAD) | SPI_WRITEREAD |

## Master(PGY-I3C-EX-PD) Request:

- 0x05-0x04-0x05-0x08-0x00-0x00-0x02-0x00-0x04-0x00-0x55-0x66

## Slave(WE2) Response:

- 0x05-0x04-0x06-0x0A-0x00-0x00-0x02-0x00-0x04-0x00-0x11-0x22-0x33-0x44

## Slave(WE2) Log:

### Slave Receive

```
[I3CS_0_SDR_Rx_Handler] size:12
i3c_slv_sdr_rx_buf[0:0x05] [1:0x04] [2:0x05] [3:0x08] [4:0x00] [5:0x00] [6:0x02] [7:0x00] [8:0x04] [9:0x00] [10:0x55] [11:0x66]
[MSG] type:0x5 Command:0x4 Flag:0x5 Payload_Len:0x0
[PACKET_TYPE_SPI]
[Need response]
[SPI_WRITEREAD]
Dump request payload
00 00 02 00 04 00 55 66 | .....Uf
BusID:0 CS_Select:0 WireMode:0 WriteSize:2 ReadSize:4
[dump i3c_slv_sdr_tx_buf]
05 04 06 0A 00 00 02 00 04 00 11 22 33 44 | .....3D
[I3CS_0_SDR_Tx_Handler] size:14
Reset TX FIFO
```

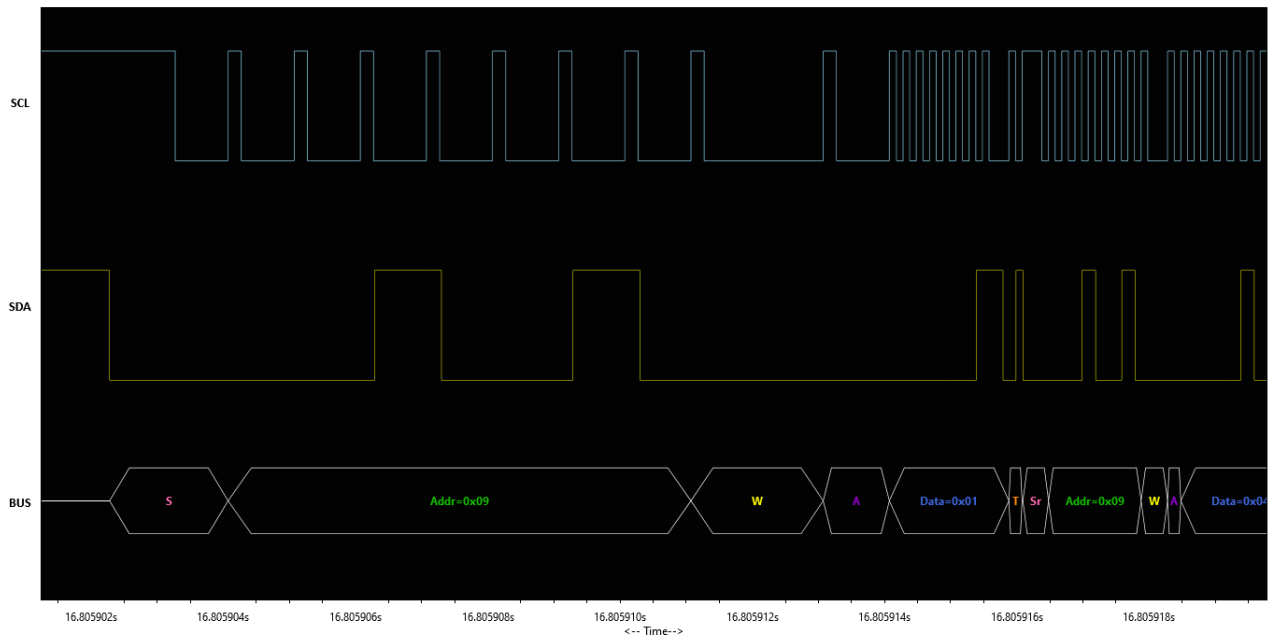
### Slave Transmit

# 4. I3C Exerciser and Protocol Analyzer

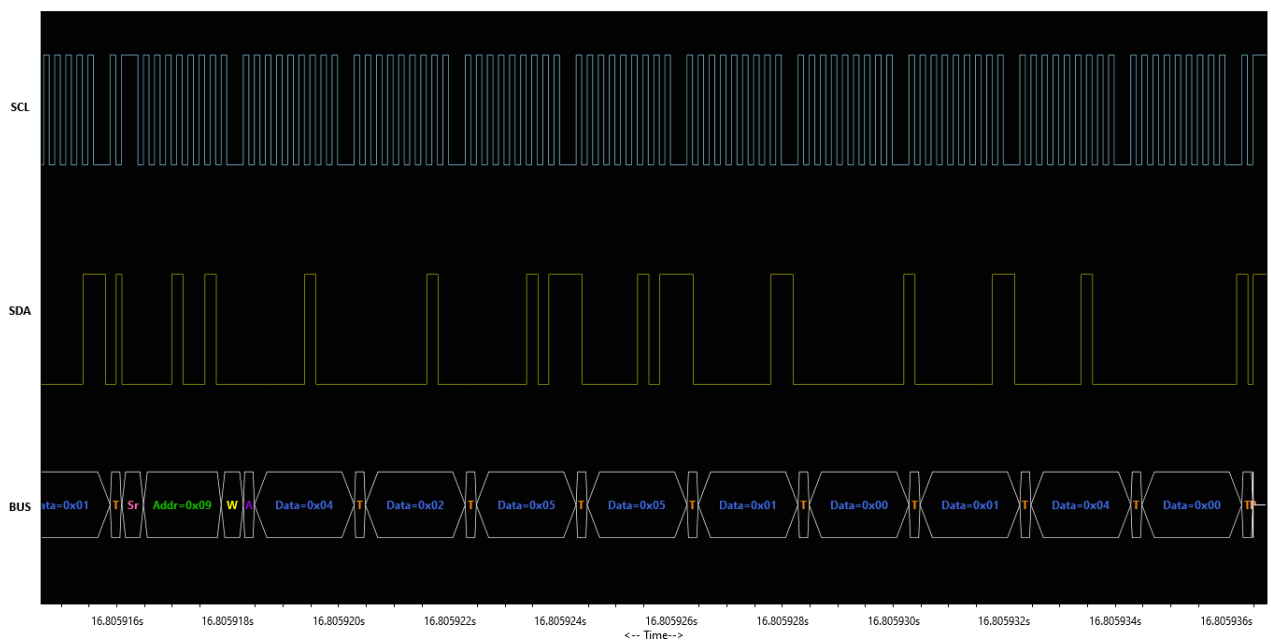
## 4.1 Command(I2C Read) Waveform Graph

|    | Frame Type      | Time       | Mode | Packet Info | Start | Addr_RW_A | CCC_T | Data_T/P | Stop | Error | Freq       |
|----|-----------------|------------|------|-------------|-------|-----------|-------|----------|------|-------|------------|
| 1. | Private_Message | 77.065483s | SDR  | Address     | S     | 0x08 W A  | -     | -        | -    | -     | 1.0001 MHz |
|    | -               | 77.065495s | SDR  | Data        | -     | -         | -     | 0x01 0   | -    | -     | 4.9505 MHz |
| 2. | Private_Message | 77.065497s | SDR  | Address     | Sr    | 0x08 W A  | -     | -        | -    | -     | 5.0000 MHz |
|    | -               | 77.065499s | SDR  | Data        | -     | -         | -     | 0x04 0   | -    | -     | 5.0001 MHz |
|    | -               | 77.065501s | SDR  | Data        | -     | -         | -     | 0x02 0   | -    | -     | 5.0001 MHz |
|    | -               | 77.065503s | SDR  | Data        | -     | -         | -     | 0x05 1   | -    | -     | 5.0000 MHz |
|    | -               | 77.065505s | SDR  | Data        | -     | -         | -     | 0x05 1   | -    | -     | 5.0001 MHz |
|    | -               | 77.065507s | SDR  | Data        | -     | -         | -     | 0x01 0   | -    | -     | 5.0000 MHz |
|    | -               | 77.065509s | SDR  | Data        | -     | -         | -     | 0x00 1   | -    | -     | 5.0001 MHz |
|    | -               | 77.065511s | SDR  | Data        | -     | -         | -     | 0x01 0   | -    | -     | 5.0001 MHz |
|    | -               | 77.065513s | SDR  | Data        | -     | -         | -     | 0x04 0   | -    | -     | 5.0000 MHz |
|    | -               | 77.065515s | SDR  | Data        | -     | -         | -     | 0x00 1   | True | -     | 5.0001 MHz |
| 3. | Private_Message | 77.165518s | SDR  | Address     | S     | 0x08 W A  | -     | -        | -    | -     | 1.0001 MHz |
|    | -               | 77.165530s | SDR  | Data        | -     | -         | -     | 0x00 1   | -    | -     | 4.9505 MHz |
| 4. | Private_Message | 77.165532s | SDR  | Address     | Sr    | 0x08 R A  | -     | -        | -    | -     | 5.0000 MHz |
|    | -               | 77.165534s | SDR  | Data        | -     | -         | -     | 0x04 1   | -    | -     | 5.0001 MHz |
|    | -               | 77.165536s | SDR  | Data        | -     | -         | -     | 0x02 1   | -    | -     | 5.0001 MHz |
|    | -               | 77.165538s | SDR  | Data        | -     | -         | -     | 0x06 1   | -    | -     | 5.0000 MHz |
|    | -               | 77.165540s | SDR  | Data        | -     | -         | -     | 0x09 1   | -    | -     | 5.0001 MHz |
|    | -               | 77.165542s | SDR  | Data        | -     | -         | -     | 0x01 1   | -    | -     | 5.0001 MHz |
|    | -               | 77.165544s | SDR  | Data        | -     | -         | -     | 0x00 1   | -    | -     | 5.0000 MHz |
|    | -               | 77.165546s | SDR  | Data        | -     | -         | -     | 0x01 1   | -    | -     | 5.0001 MHz |
|    | -               | 77.165548s | SDR  | Data        | -     | -         | -     | 0x04 1   | -    | -     | 5.0000 MHz |
|    | -               | 77.165550s | SDR  | Data        | -     | -         | -     | 0x00 1   | -    | -     | 5.0001 MHz |
|    | -               | 77.165552s | SDR  | Data        | -     | -         | -     | 0x0C 1   | -    | -     | 5.0001 MHz |
|    | -               | 77.165554s | SDR  | Data        | -     | -         | -     | 0x00 1   | -    | -     | 5.0000 MHz |
|    | -               | 77.165556s | SDR  | Data        | -     | -         | -     | 0x38 1   | -    | -     | 5.0001 MHz |
|    | -               | 77.165558s | SDR  | Data        | -     | -         | -     | 0x85 0   | True | -     | 5.0001 MHz |

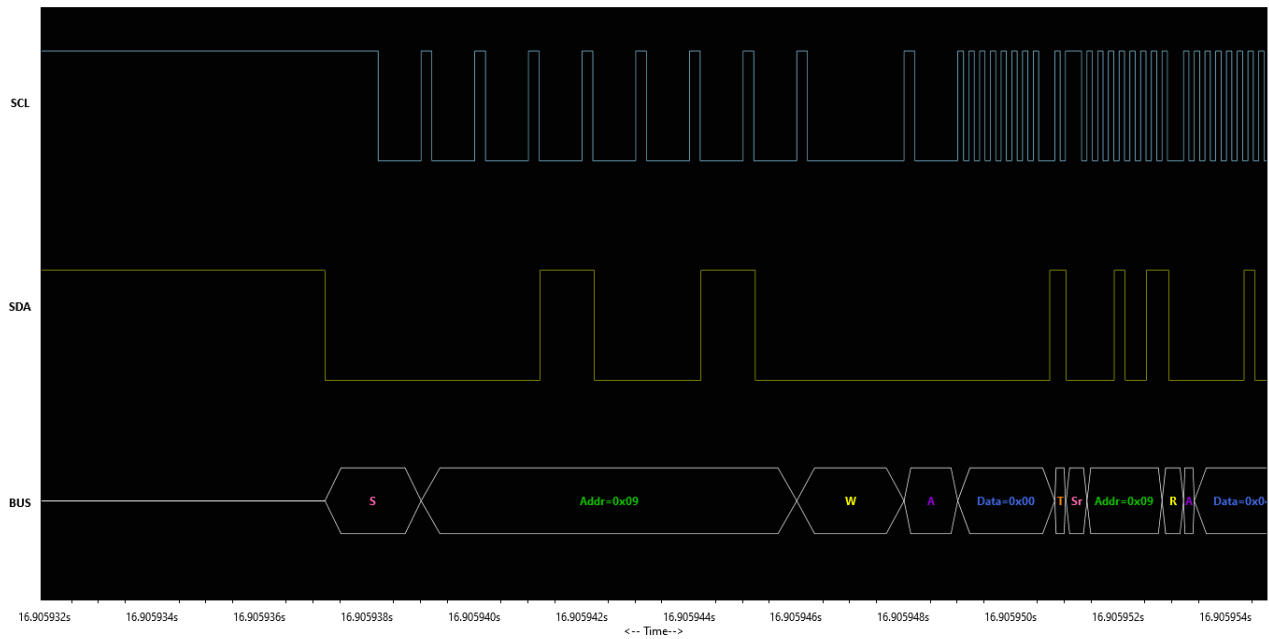
1. Master write reg\_addr 0x01(SDR\_RX\_FIFO).



## 2. Master write data(0x04-0x02-0x05-0x05-0x01-0x00-0x01-0x04-0x00).



## 3. Master write reg\_addr 0x00(SDR\_TX\_FIFO).



#### 4. Master read 13 bytes

**data(0x04-0x02-0x06-0x09-0x01-0x00-0x01-0x04-0x00-0x0C-0x00-0x38-0x85).**

